

## PINN 2D

$$\text{PDE: } \frac{\partial u}{\partial t} = \alpha \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \text{ on } [0, 1]^2 \times [0, T]$$

**2D PINN with Hard Boundary Conditions**  
 $u_{xx} + u_{yy} = f$ ,  $u = \sin(\pi x)\sin(\pi y)$

